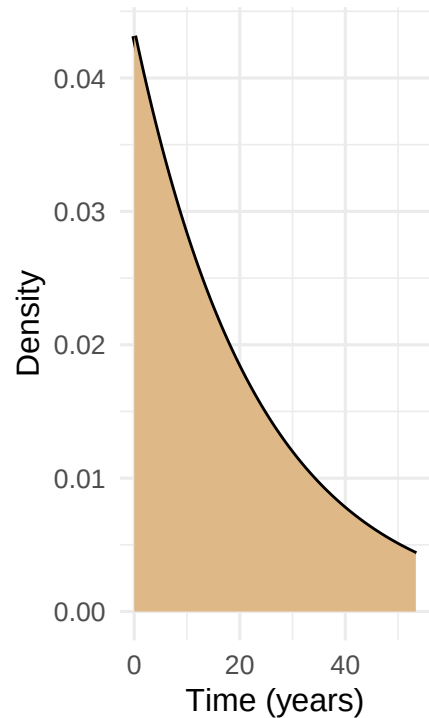
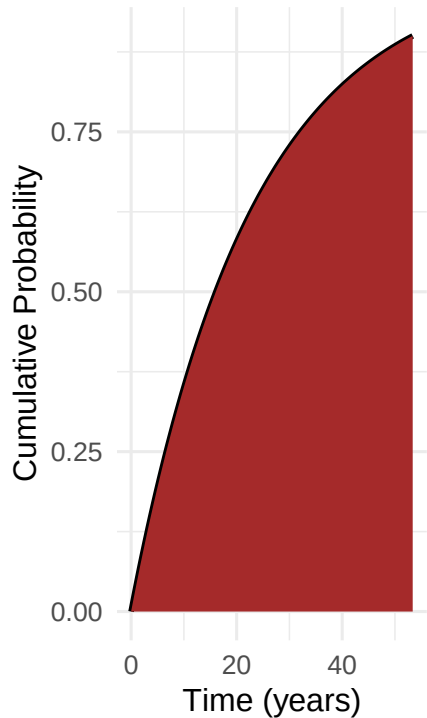


Solar Panel Reliability Report

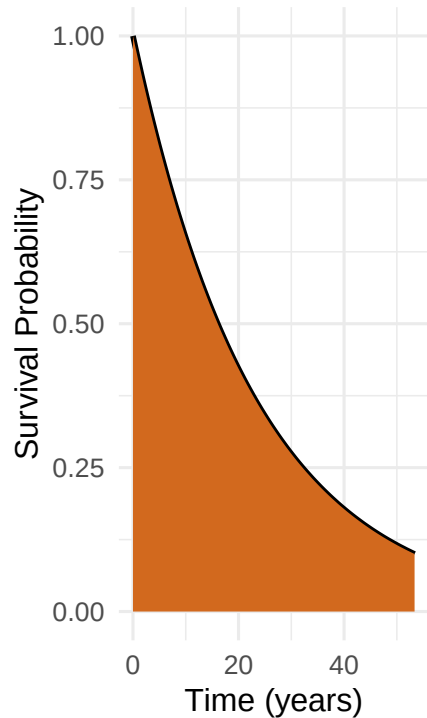
PDF



CDF



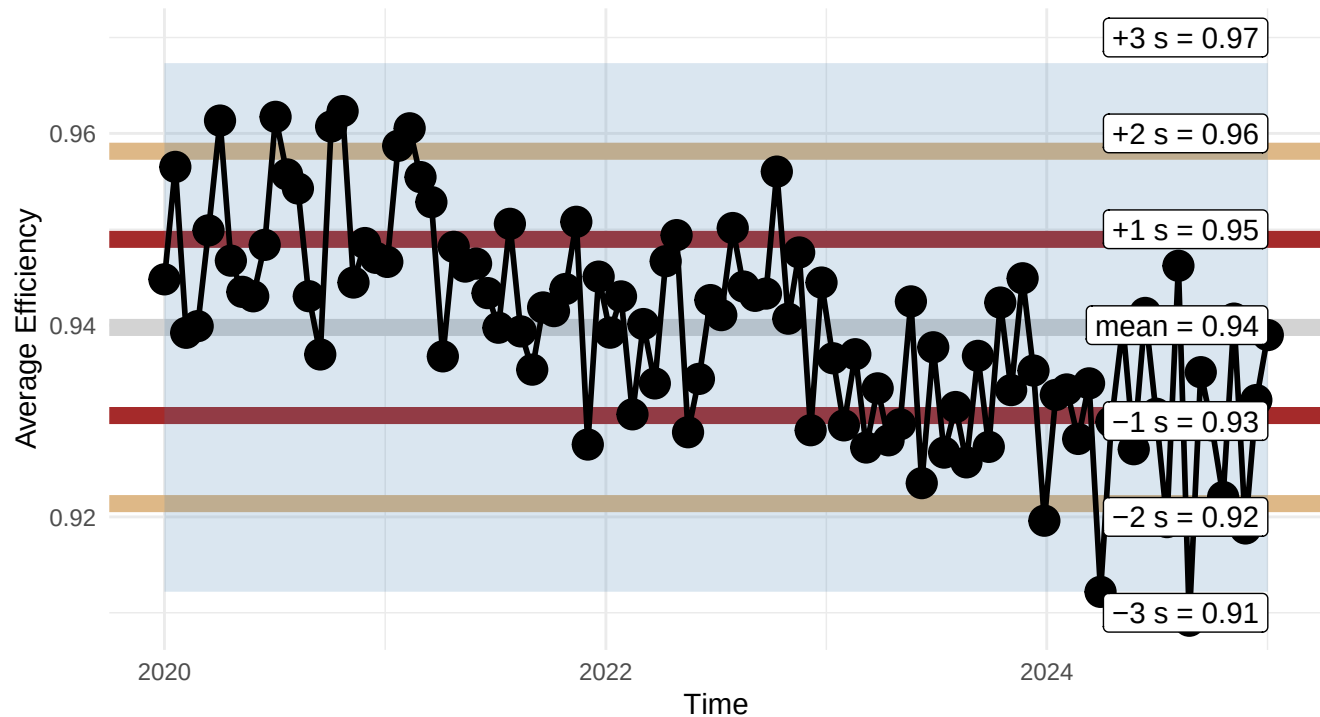
Survival Function



Reliability was modeled using an exponential lifetime distribution. Estimated MTTF =23.2 years. Based on 100 panels and 21 observed failures, the expected survival rate is 65.0% after 10 years and 34.0% after 25 years. The slope of the control chart indicates a decreasing trend, suggesting decreasing efficiency over time. Most of your failures are due to wiring issues. Contact electricians.

Metric	Value
Total panels	100
Failed panels	21
MTTF (years)	23.2
Failure rate (/yr)	0.0432
Survival @10y	65.0%
Survival @20y	42.2%
Survival @25y	34.0%

Control Chart



Number of Solar Panels Failed by Failure Type

